MODEL-ZOO / HEAVY-TAILED SELF-REGULARIZATION

Model weights as evidence

Exploratory audit: weight grouping, base retrieval, and performance controls



10,946

lineage models

Phase 2 + ablation

17

weight-derived groups

selected without labels

3.21

non-alpha modern MAE

near-group held-out audit

September 2026 results / October 2026 scale-up plan

Exploratory reassessment: reassessment_0063666fd5733b453c38

Keep predictor inputs separate from evaluation labels

Unlabeled discovery

INPUT
complete core + calibrated partial roles

→

FROZEN OUTPUT
clusters + representative models

Known base candidates

INPUT
profiles + fixed candidate list

→

FROZEN OUTPUT
ranked candidate bases

Performance

INPUT
matrix summaries + training scores

→

FROZEN OUTPUT
held-out score estimates

EVALUATION ONLY | declared source roots and family labels score frozen lineage predictions; held-out benchmark scores score predictions

Collection metadata shaped selection; collection IDs define one holdout. IDs join records and break ties, not predictor features. This is exploratory reuse, not untouched confirmation.

Sources: reassessment_0063666fd5733b453c38/protocol.json; parent prediction freezes

Four phases, each ending in a checked output

1 Assemble model lists

Phase 2 and ablation IDs canonicalize and remove duplicates

19,713 selected records

2 Analyze weights

read checkpoint matrices compute alpha by layer and role

10,946 lineage / 11,801 performance

3 Make predictions

form groups · rank known bases estimate benchmark scores

predictions saved before labels

4 Validate

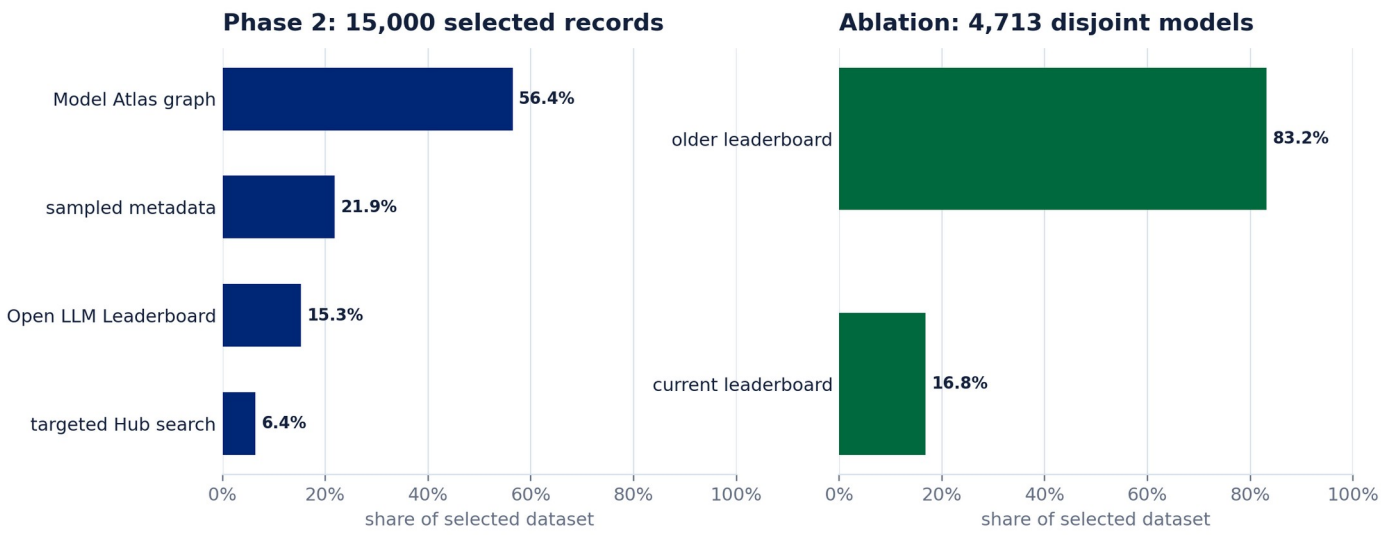
score against declared source links measure held-out benchmark error

reported results and limits

Predictor features come from saved weights; score fitting uses training outcomes. Collection metadata shaped curation and one holdout, so the full process is not metadata-free.

Sources: collection code, parent weight-only run, and exploratory reassessment protocol.

Two collections were built for different jobs



Collection metadata shapes coverage. Predictor features exclude these labels, but source selection and joins remain part of the evidence boundary.

PHASE 2 · BROAD MODEL ZOO

1 Combine the Model Atlas graph, sampled records, leaderboard rows, and targeted Hub searches.

2 Normalize repository IDs and declared revision fields; the analyzed checkpoint revision is not verified.

3 Fill model-type quotas while balancing score and lineage availability; favor rows with existing weight results.

ABLATION · DISJOINT SCORE TEST

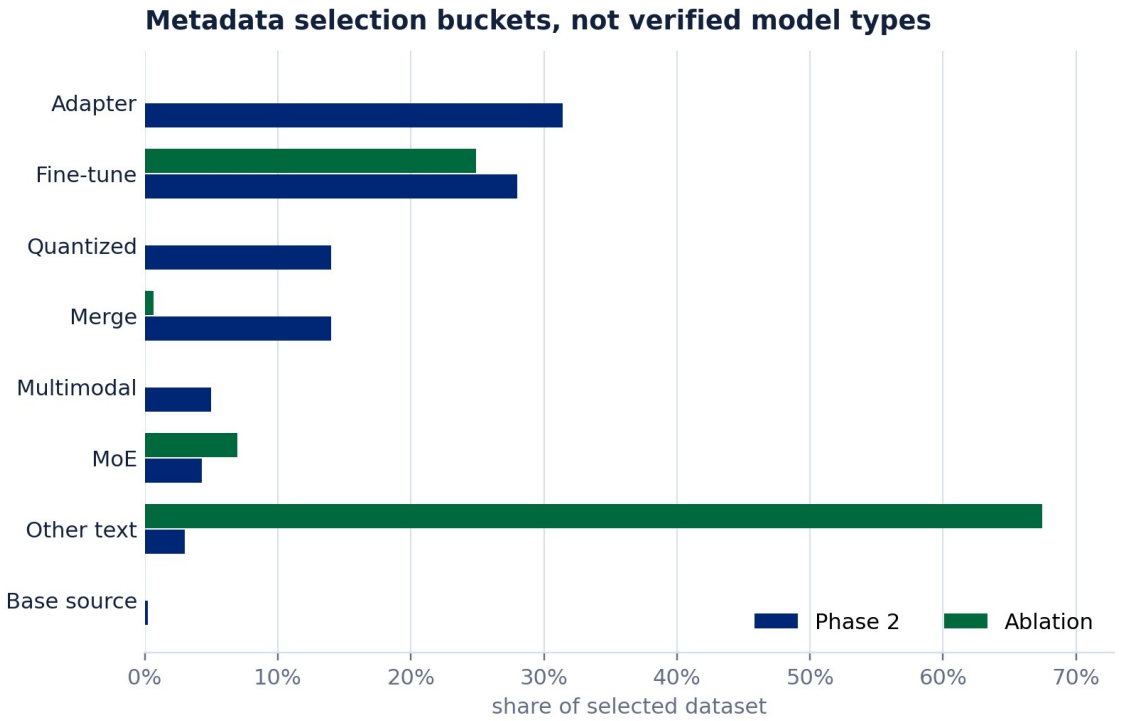
11,836 leaderboard candidates → **4,713 selected**

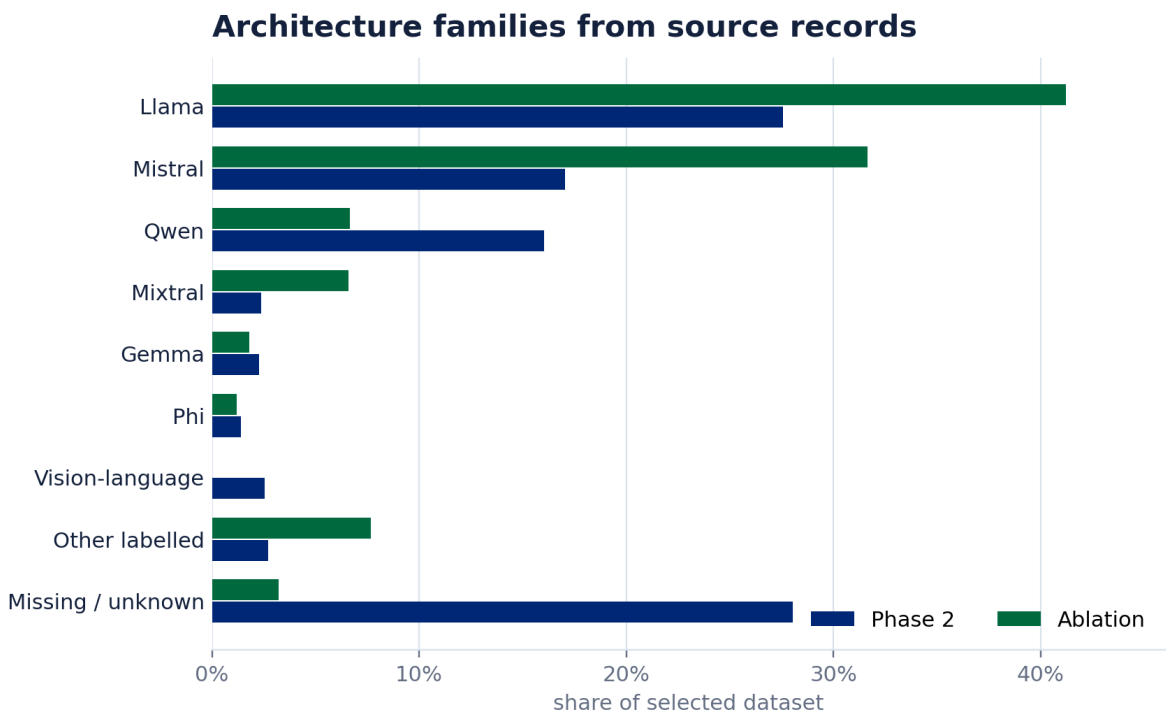
Removed: 4,177 Phase 2 overlaps · 2,674 inaccessible repositories · 272 lower-priority duplicates

Keep score coverage high, rotate across model types, cap missing scores at 15%, and stop below 5,000 when no more rows pass.

Sources: fetch_data/build_phase2_zoo.py, fetch_data/build_ablation_dataset.py, and ablation_source_manifest.json

Selection buckets describe curation; ablation is benchmark-heavy





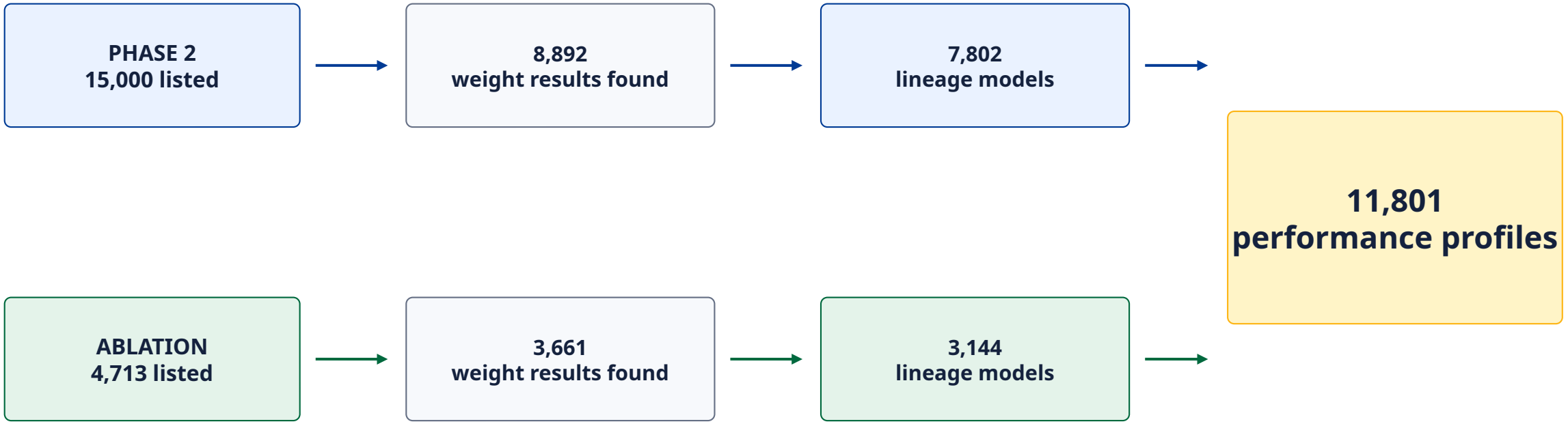
PHASE 2 · 15,000
2,922 average-score records · 13,301 declared source links
10,793 architecture labels · 2,284 parameter counts

ABLATION · 4,713
4,662 average-score records · 291 declared source links
4,561 architecture labels · 4,349 parameter counts

Type-chart categories are selection buckets, not verified model types. Source links are declared, not verified ancestry; architecture labels are descriptive.

Sources: completed run's Phase 2 and ablation source tables; architecture graph used descriptively after selection.

Shared weight checks lead to task-specific coverage

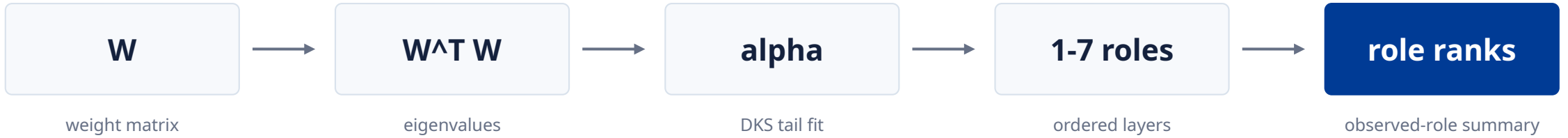


Strict seven-role core: 10,431 models. Reliable missing-role patterns add 515 lineage assignments. Performance uses the label-free cutoff of at least 4 observed roles.

These are parent role-profile counts. The reassessment also uses all-matrix summaries; source IDs define the collection holdout, not predictor features.

Sources: weight_only_2a5abfeb98dc33d9662c/population_ledger.csv; source-run provenance.json and exclusions.csv

Compare models using the layer roles that are actually observed



Attention: query · key · value · output

Feed-forward: gate · up · down

THIS STUDY

Custom stored DKS alpha, not a WeightWatcher run · depth scaled to [0,1] · 32 positions per observed role · within-role ranks for grouping

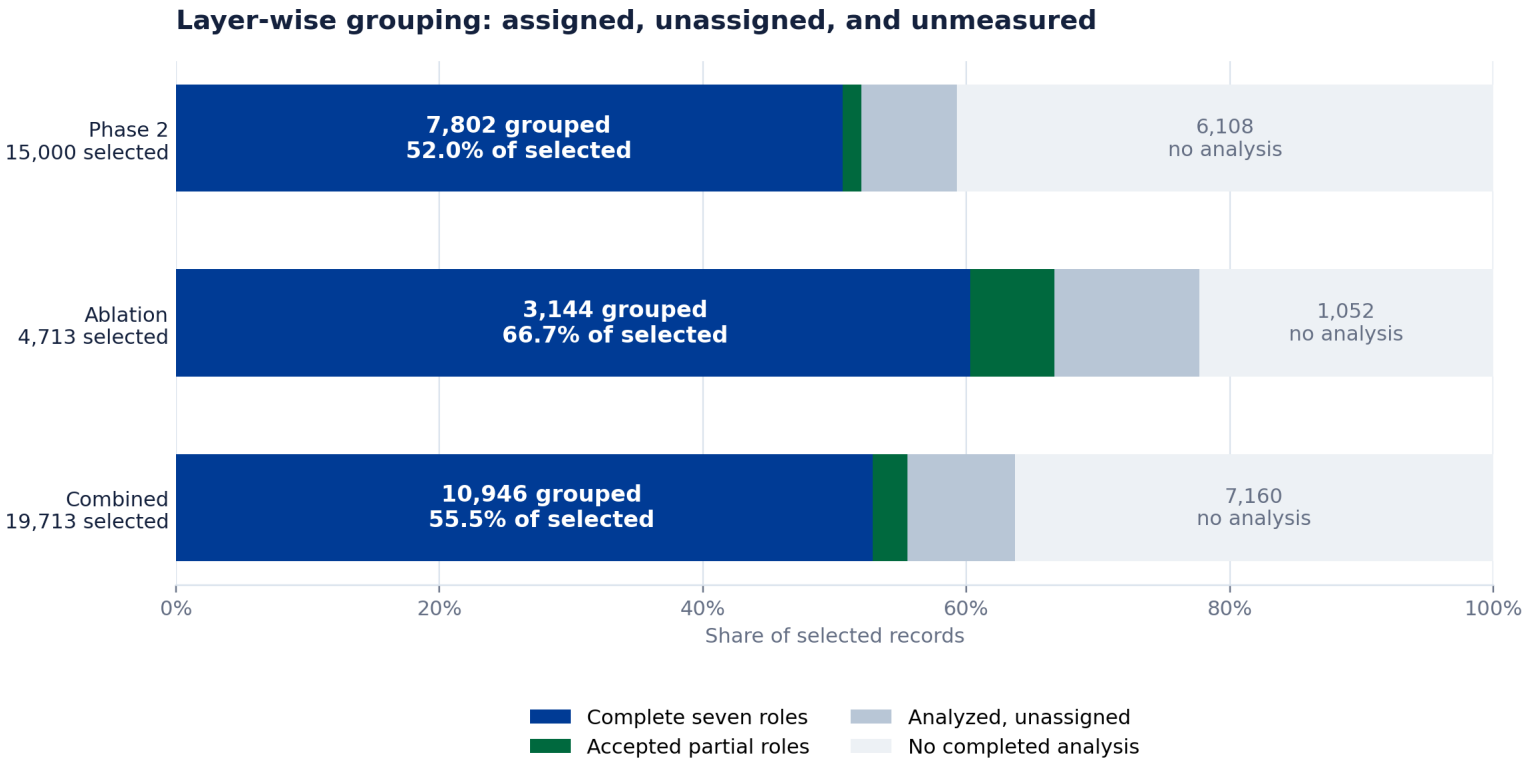
REFERENCE PAPER

Fixed-tail Hill alpha, k=n/2 · raw-alpha clustering · role-wise Spearman similarity · order-preserving DP for unequal depths · MoE experts averaged

Ranks discard alpha offsets, scales, and spacing, but retain order. Absolute alpha and other saved matrix statistics can still carry performance signal.

Sources: stored DKS alpha values by layer; archive/reference-work/Spectral Signatures of Large Language Models.html

How much of the collection can layer-wise grouping cover?



10,946

grouped models

87.2%

of analyzed models

10,946 / 12,553

55.5%

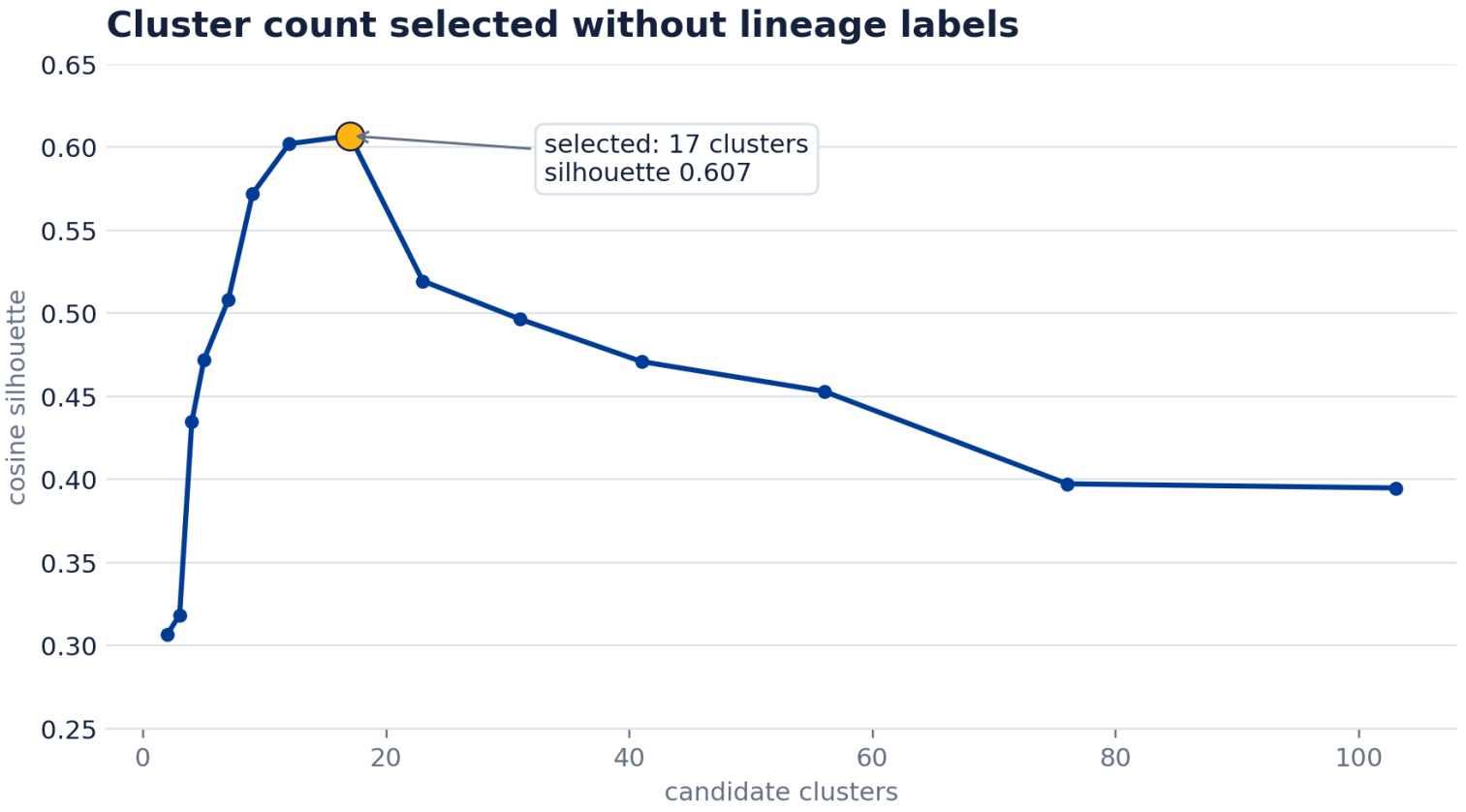
of selected records

10,946 / 19,713

Fixed seven-role vocabulary; only tested missing-role patterns are admitted. 7,160 selected records have no completed analysis. Unseen-architecture accuracy remains untested.

Sources: weight_only_2a5abfeb98dc33d9662c/population_ledger.csv; retained combined-study provenance.json

17 core groups give the clearest separation without family labels



0.607

group separation

higher is better

0.944

repeat-run agreement

average of 3 reruns

0.883

lowest rerun agreement

1.541

within/between spread

lower is better

HOW 17 WAS CHOSEN

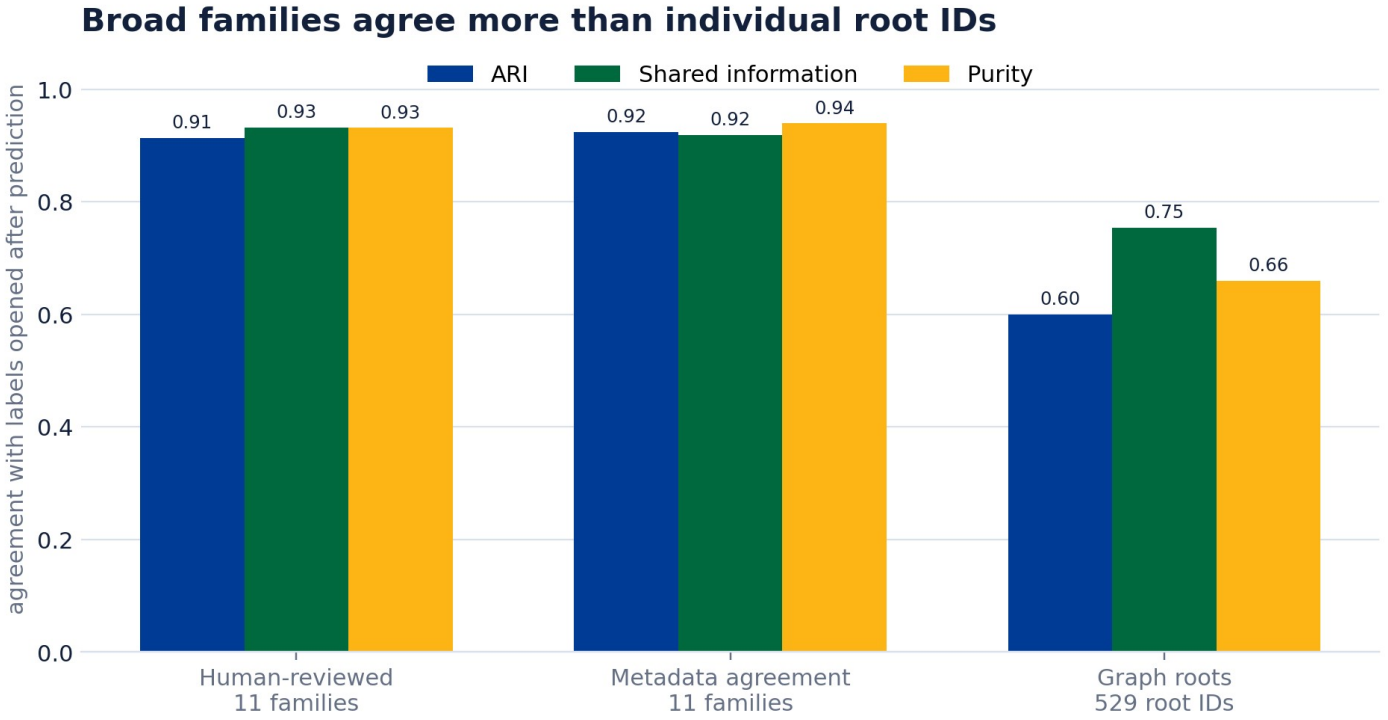
Fit candidate counts on 10,431 complete profiles

Keep the count with the best separation

515 partial profiles are attached only after label-free masking checks pass.

Sources: weight_only_2a5abfeb98dc33d9662c/lineage_cluster_selection.csv and lineage_internal_metrics.csv

Validate grouping against overlapping, imperfect label views



LABELS OPENED AFTER GROUPING

Human-reviewed

174 models assigned to 11 broad base families by manual review.

Metadata agreement

2,981 models assigned to 11 broad families when metadata rules agree.

Declared roots

5,013 models linked to 529 graph-component roots; not verified historical ancestry.

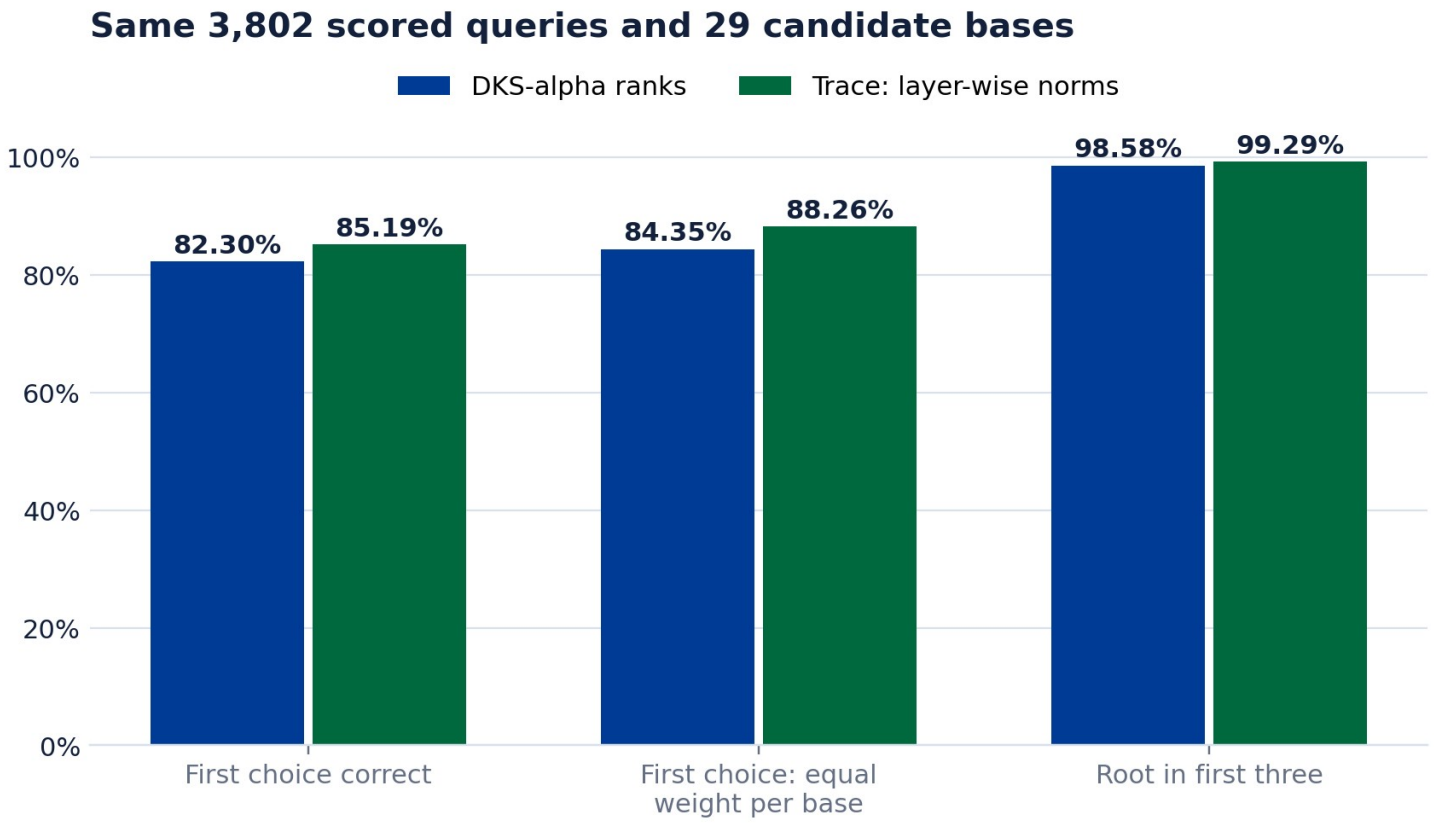
ARI
Pairwise grouping agreement, corrected for chance. 1 means identical grouping.

Shared information
How much knowing one grouping tells us about the other. 1 is complete.

Purity
Majority-label share inside each predicted group. It can favor too many small groups.

Sources: parent lineage_validation_metrics.csv; reassessment_0063666fd5733b453c38/all_matrix_cluster_validation.csv

Base matching depends on the comparison and treatment of ties



29 / 38
available / supplied bases

3,802
matched known-root queries

82.30%
rank top-1

85.19%
Trace top-1

Trace compares weight magnitude through layers, averaging seven role-wise correlations. Its first-choice gain is sensitive to tied candidates; the full and tie-excluded subsets answer different questions.

757 other labeled queries have a root absent from the supplied list; 278 have a listed root without usable measurements. These are queries, not additional candidate bases.

After excluding exact ties in either method: rank 85.88%, Trace 84.73%. This changes the model/root mix; equal-weight-per-root accuracy still favors Trace.

Sources: reassessment_0063666fd5733b453c38/lineage_gallery_metrics.csv; audit/gallery_tie_sensitivity.csv. Trace: arxiv.org/html/2608.07786v1

Our larger study is not an exact replication

Dimension	Spectral Signatures paper	Exploratory reassessment
Prediction boundary	Weight similarities; training scores for prediction	Weight features; training-only score fitting
Alpha estimator	Fixed-tail Hill, k = n/2	Custom stored DKS; not WeightWatcher or fixed Hill
Grouping / similarity	Flattened raw alpha / role-wise Spearman	Rank grouping retained; separate Trace base test
Unequal depth	Z-score + order-preserving matching	Ranks: 32-point interpolation; Trace: pair-max depth
Performance setup	Five folds stated; 100 half-splits in results	Five fixed near-group parts; explicit stress splits
Scope of comparison	Different candidates, cohorts and targets	No matched replication; no claim to beat the paper

The paper's split-description discrepancy is unresolved. Neither its accuracy nor our MAE establishes superiority across different populations and protocols.

Sources: arxiv.org/html/2607.03377v1, Sec. 4.2 and 5.3; reassessment_0063666fd5733b453c38/protocol.json

Keep similar models together in five test parts

BUILD FIVE TEST PARTS

Connect near profiles similarity >= 0.995 → Keep each group intact in five fixed parts → Freeze the same splits for predictor contrasts

REPEAT FIVE TIMES

Hold out 1 part → Use the other 4 parts → Fit preprocessing on training rows only → Fit fixed trees / ridge or rank 3-neighbor model → Open the held-out score and measure error

Fixed trees: 200 iterations, 15 leaves, rate 0.05, L2=10, minimum leaf=20. Ridge penalty=100; no held-out tuning.

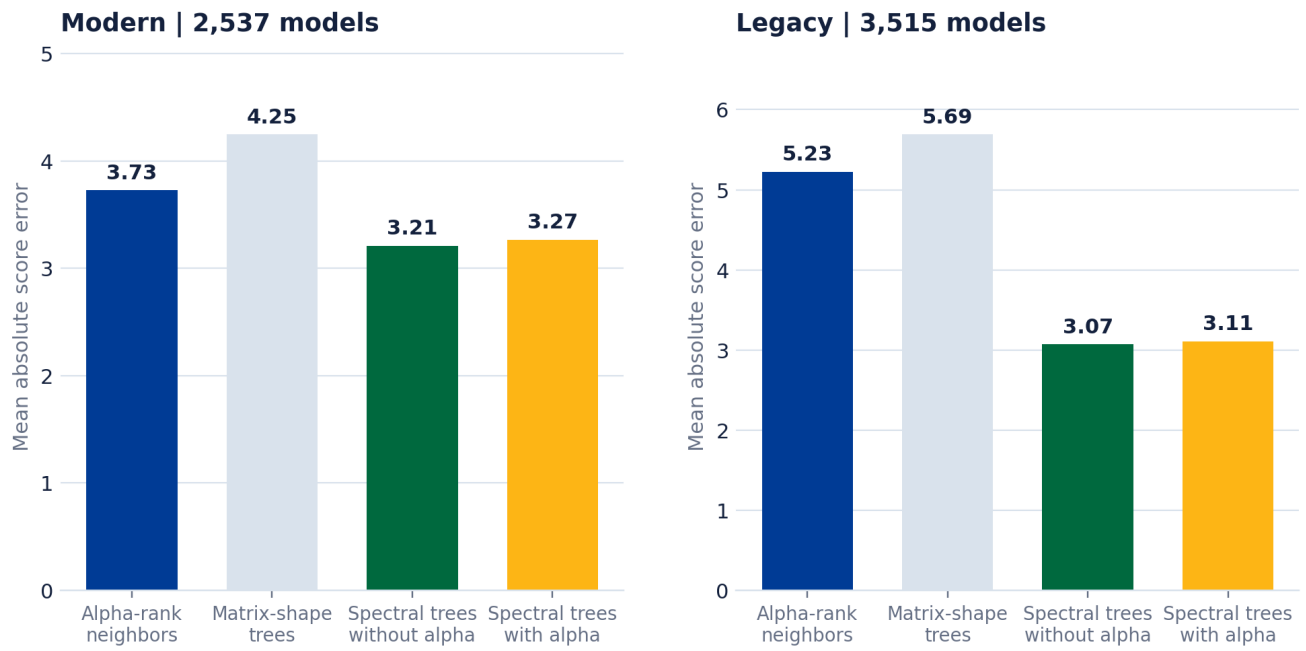
Controls: shape, non-alpha spectra, combined spectra, and rank neighbors. Average-score targets and individual tasks are evaluated separately.

Training scores are permitted; held-out scores are for evaluation. Collection labels define one stress split. Previously studied data: exploratory, not untouched confirmation.

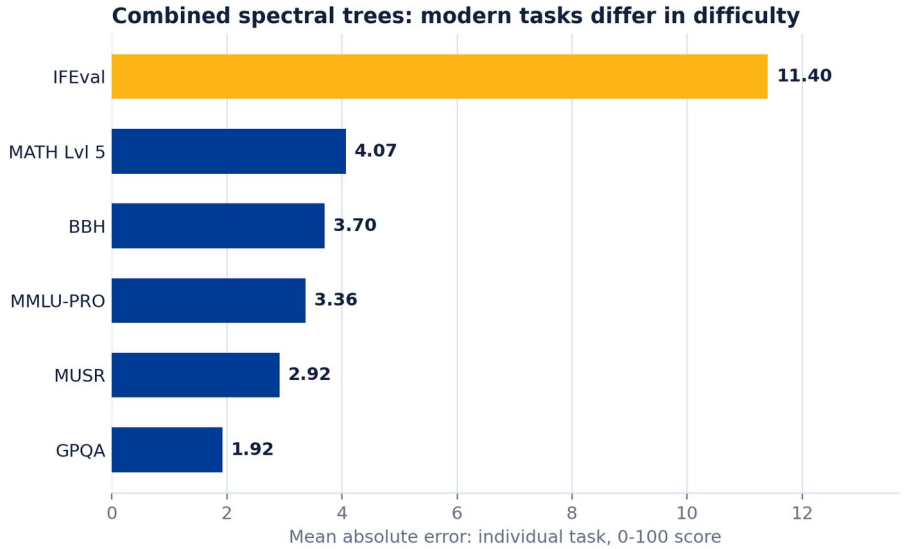
Sources: reassessment_0063666fd5733b453c38/protocol.json, splits.csv and performance_predictions.csv

Alpha does not improve the key performance controls

Near-matches kept together: adding alpha does not improve these trees



Shape 4.249 / 5.693 · rank 3.726 / 5.227
2,537 modern / 3,515 legacy rows; same near-group splits.



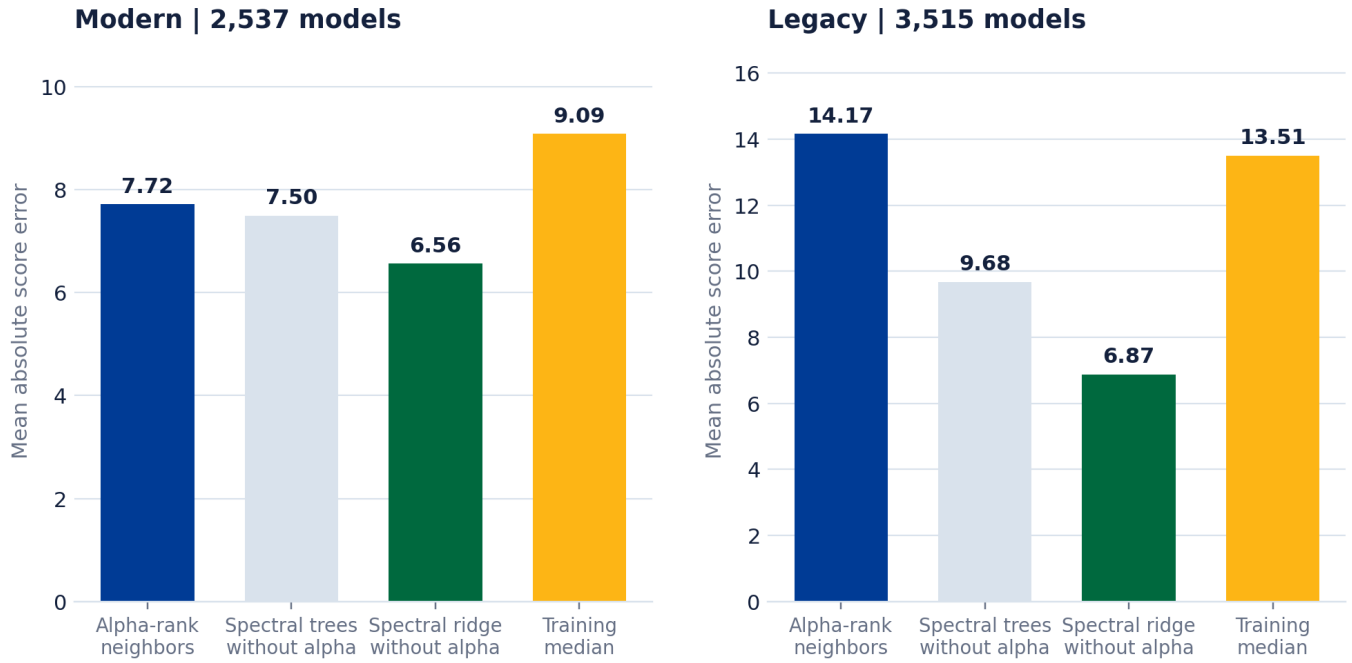
AVERAGE-SCORE MAE: MODERN / LEGACY
Non-alpha trees 3.211 / 3.068
Combined trees 3.266 / 3.109

Alpha-specific gain is not established. Combined = alpha plus other matrix measurements; individual-task checks do not replace the controls.

Sources: reassessment_0063666fd5733b453c38/performance_metrics.csv and performance_task_metrics.csv; near_group_0995, matched_role_population

Unseen groups and collections expose larger errors

An unseen weight group raises errors; linear spectral information still helps



NON-ALPHA RIDGE · WHOLE-GROUP MAE

6.561
modern

6.873
legacy

Whole weight clusters are held out together. This is a shift test, not the near-group interpolation test; compare each method with the training median.

COLLECTION-HOLDOUT WARNING
Combined ridge produces 26 modern predictions outside 0-100. These are range failures, not plausible benchmark forecasts.

Errors change across held-out groups and collections. Inspect the score range and coverage, not only average error.

Sources: reassessment_0063666fd5733b453c38/performance_metrics.csv; audit prediction_metrics_audit.csv and collection_direction_metrics.csv

Keep rank grouping, test Trace bases, and retain non-alpha controls

Retain

- Keep layer-wise grouping: all-matrix summaries cover more models but separate families poorly
- Trace as a base-retrieval comparison, with fixed gallery and tie diagnostics
- Non-alpha trees as baseline: MAE 3.211 modern / 3.068 legacy

Interpretation

- Ranks discard magnitude; direct measurements predict scores better
- Adding alpha did not improve the key spectral controls
- Local prediction and broad grouping are distinct from historical ancestry

Remaining uncertainty

- Repository-ID source joins do not verify the analyzed checkpoint revision
- Missing bases and near-tied candidates limit exact-ID retrieval
- Whole-group shifts worsen errors; collection-shift ridge can leave 0-100

Next experiment: use the newer pinned collection to test whether weight measurements improve task-specific model choices. See the October scale-up plan in the appendix.

Sources: reassessment_0063666fd5733b453c38/protocol.json, audit tables, and frozen lineage evaluation

APPENDIX

Methods and next study

Operational details, next weight-based methods, and terms used in the main results

A1 — Collection and GPU scheduling

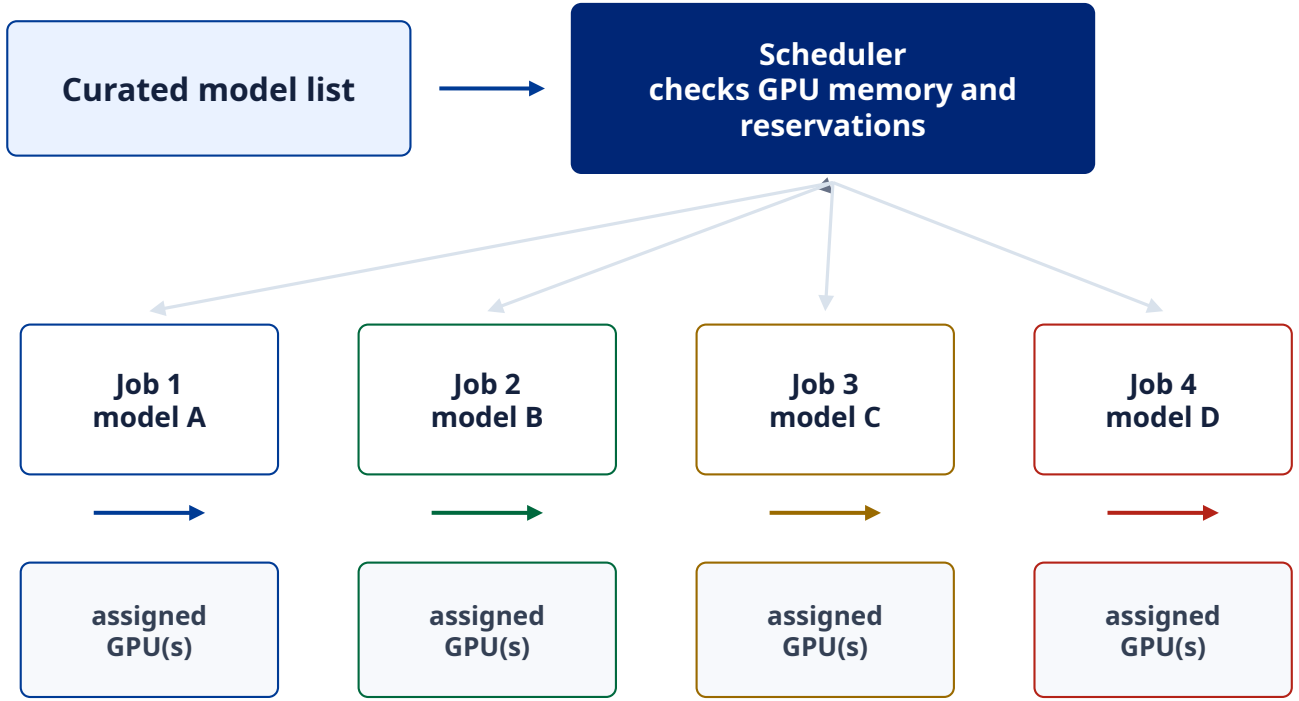
A2 — Matrix analysis and storage

A3 — Future direct weight comparison

A4 — Terms and evidence guide

A5 — October scale-up research plan

One scheduler assigns jobs to available graphics processors



CONFIGURABLE CONTROLS

- which GPU IDs may be used
- GPUs per job and jobs at once
- memory threshold before assignment
- retry count and stale-job timeout
- reload, drain, or stop while running

Each job uses its own download cache. Removing it after the job prevents storage growth, at the cost of downloading shared files again.

The memory check and internal reservation avoid overlap among these jobs; they are not a cluster-wide lock against unrelated users.

Historical collection workflow; newer operational collector: /jumbo/yaoqingyang/kinshuk/Model-Zoo/docs/operations/analysis.md

From weight matrices to saved spectral measurements



PLUG-IN ANALYSIS CHOICES

DKS
choose the tail start that best fits

Fixed rules
use a specified part of the tail

STRUCTURED OUTPUTS

CSV / HDF5
per-layer metrics and model fields

Full spectra
current collector: saved by default; opt-out available

Logs and status
progress, failures, retries, and timing

Matrix work can be split across several GPUs. This lowers wall-clock time when matrix operations dominate, but does not guarantee fewer total GPU-seconds.

New collector: pinned revisions and matrix coverage. Historical results here still use stored DKS alpha; the new default uses the fixed-tail rule.

Sources: historical model statistics; current collector: /jumbo/yaoqingyang/kinshuk/Model-Zoo/docs/operations/analysis.md

Add direct weight-matrix comparisons for within-family verification

STAGE 1
rank grouping + Trace retrieval
10,946 rank assignments

→

STAGE 2
direct weight-matrix check
small candidate set

→

OUTPUT
relatedness evidence
calibrate on separate pairs

arXiv:2608.07786

Spectral energy + subspace alignment

Trace tested from sqrt(norm). Fine directional comparison requires top singular vectors, not scalar summaries.

arXiv:2608.14929

Centered residual signatures

Center Wout Win by subtracting trace(M) I/d. Requires compatible blocks; symmetric similarity is not ancestry direction.

AWM · arXiv:2510.06738

Assignment + centered alignment

Embedding assignment and sign alignment, then Q/K unbiased CKA. Requires direct weights; not tested in this audit.

Trace is feasible from saved scalar norms and was tested. Directional, residual, and AWM checks still require unavailable direct-weight information; none alone proves historical ancestry.

Primary sources: https://arxiv.org/html/2608.07786v1; https://arxiv.org/html/2608.14929v1; https://arxiv.org/html/2510.06738v3

Measures used in this study

Alpha — Exponent fitted to the upper eigenvalue distribution of a weight matrix.

DKS — Chooses the tail start that minimizes the Kolmogorov-Smirnov fit distance.

ARI — Adjusted Rand Index: grouping agreement corrected for chance; 1 is exact.

Spearman — Agreement in rank order; 1 means the two rankings match.

MAE — Mean absolute error: the average size of a score prediction error.

Near-group test — Keep near-profile groups intact in five fixed parts; fit scores using the other four.

Trace — Mean role-wise Pearson of Frobenius depth profiles; resample to pair-max depth.

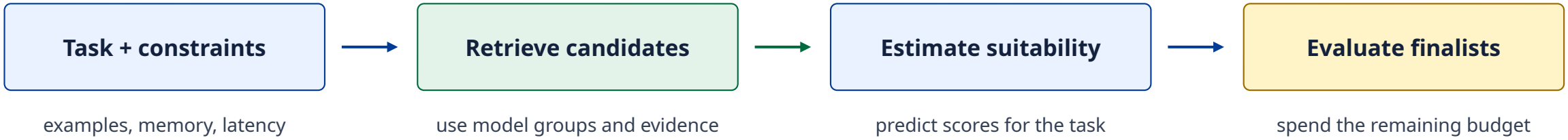
Representative — The observed core model closest to its group mean; not necessarily its historical base.

Evidence boundary: saved weight features plus training scores form predictions. Declared source links evaluate frozen retrieval; collection metadata shaped selection and one holdout. Checkpoint revision remains unverified.

Definitions follow the analysis code and cited reference methods.

Do weight measurements improve model selection for a task?

Measure score lost versus the best feasible model, evaluation cost, and constraint failures.



COLLECTOR CAPABILITIES NOW

Pinned revisions, canonical matrix records, full computed spectra, and explicit coverage.
New records still need verified benchmark joins and prediction experiments.

PROPOSED EVIDENCE COMPARISON

Scientific track: weights plus training scores.
Practical track: documentation, historical scores, and weights.
Test new roots, tasks, and future releases.

PROPOSED COLLECTION MILESTONES

1k pinned artifacts

Measure collection cost and coverage

5k-10k measured

Compare task choices and evidence

50k measured target

Release the zoo and benchmark

Source: docs/research/scaleup/model_zoo_frontier_research_plan_2026-10-23.md; proposed experiments and milestones.